Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Turek, Fred W Ph.D.		
Project Title:	Animal Model for Sleep Loss and Circadian Disruption		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60201-2919	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	04/01/2001	**End Date:**	07/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Reid, Kathryn (Northwestern University)		
Grant/Contract No.:	HPF.002.06		
Performance Goal No.:			
Performance Goal Text:			
	The adverse effects associated with imposed disruptions of the normal circadian and sleep-wake cycles are particularly relevant to NASA personnel and their ability to carry out normal duties at a high level of efficiency. Many space travel situations demand that personnel both ground-based and in flight engage in duty schedules that can lead to circadian rhythm disruption and sleep loss. The tasks that can be affected involve vigilance, operation and control of vehicles/aircraft, maintenance, preparation and operation of equipment, as well as command and control activities. Night operations are important for successful missions, and there is a clear need to find countermeasures that can alleviate the adverse effects of these activities on human circadian rhythms and sleep as well as on neurobehavioural capabilities and on physical performance. We have recently developed an animal model for chronic circadian disruption and partial sleep loss in rodents, including rats, hamsters and mice, by inducing daily periods of imposed wakefulness at different times of day that result in alterations in the distribution of sleep across the 24-hour day and to curtailment of the amount of sleep per day. In the		

Task Description:	proposed studies we will use the mouse as our animal model of choice because of the genetic resources available in this species, and because of our ultimate goal of being able to utilize genetic approaches for alleviating the adverse effects associated with chronic sleep loss and circadian disruption. Use of this model will allow us to determine the effects of chronic misalignment of the activity-rest cycle with both the normal circadian time and the entraining light-dark cycle, and to develop countermeasures to the adverse effects associated with chronic sleep loss and circadian disruption. The specific aims of the studies to be carried out with this new animal model are: 1. To determine the effects of 12 hours of imposed wakefulness during both the normal active and inactive periods on circadian rhythms, the sleep-wake cycle and neurobehavioural and motor performance measurements. 2. To test the hypothesis that treatment with either a physiological or pharmacological dose of melatonin at the beginning of the imposed period of wakefulness will alter the effects of this temporal desynchrony on the circadian clock, the sleep-wake cycle, and/or on neurobehavioural and motor performance measurements. 3. To test the hypothesis that access to a wheel (exercise) when in the home cage, will alter the effects of the imposed periods of wakefulness on the circadian clock, the sleep-wake cycle, and/or neurobehavioural and motor performance measurements.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An overall objective of this project is to integrate the findings from the proposed studies with research being conducted by other NSBRI team members working in humans. The development of an animal model for disrupted rhythmicity, sleep loss, and the misalignment of the sleep and circadian clock system is expected to yield new insights and countermeasures for the treatment of sleep and circadian disorders that will benefit humans involved in extended missions in space as well as for those with disrupted temporal organization under non-space conditions. These countermeasures could be useful in a number of situations involving NASA personnel, particularly in extended duration space flight missions that will result in challenges to the sleep and circadian system of the flight crew and support teams. In addition these same countermeasures would be useful in circumstances where people are exposed to circadian disruption and sleep loss, for example shiftworkers.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 07/16/2025)